

Bridging the Measurement Gap in Augmented Intelligence: The Human Enhancement Quotient (HEQ) and Augmented Intelligence Score (AIS)

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SSRN Abstract ID: 6195238

April 2026

Abstract

Despite \$2.9 trillion in projected economic value from human-AI collaboration (McKinsey Global Institute, 2025) and widespread organizational investment in AI-augmented workflows, no cross-platform, behavior-anchored, governance-integrated instrument identified in the present review has achieved broad adoption for measuring how well a specific individual performs within the augmented intelligence partnership. This paper presents the practitioner-research pathway that produced the Human Enhancement Quotient (HEQ) and Augmented Intelligence Score (AIS) as responses to this measurement gap, situating both instruments within the proposed measurement science framework of Human-AI Collaborative Intelligence (HACI). The paper documents: (a) the evidentiary methodology that preceded AI adoption and informed instrument design; (b) the practitioner discoveries, including fabricated AI outputs, multi-platform disagreement patterns, and reverse automation bias, that generated the governance architecture; (c) the four-dimension instrument structure pilot-tested across five AI platforms with a cross-platform consistency coefficient ($ICC = 0.96$) computed via the parallel Factics Intelligence Dashboard; (d) the competitive environment of adjacent instruments as of April 2026; and (e) the integration of checkpoint governance audit trails as supplementary quantitative evidence of human contribution. Preliminary cross-user testing ($n=10$) suggests dimensional variance within four points and consistent identification of Collaborative Intelligence Quotient as the lowest-scoring dimension, consistent with findings by Ng Lane (2025) on the dangers of deferring to AI and Vaccaro et al. (2024) on conditions under which human-AI combinations underperform. The paper does not claim universal psychometric validity. It presents an instrument, an initial evidence base, a staged validation roadmap, and an open invitation for replication. All frameworks are published open-source under Creative Commons (github.com/basilpuglisi/HAIA).

1. Introduction: The Measurement Gap

Gartner (2019) defined augmented intelligence as a human-centered partnership model of people and artificial intelligence working together to enhance cognitive performance. Organizations responded with investment at scale. What did not arrive alongside the investment was a way to measure whether the collaboration was actually working for a specific individual.

The consequences of this gap are now empirically visible. The World Economic Forum's Future of Jobs Report 2025 projects 92 million jobs displaced by 2030 and 170 million created, for a net addition of 78 million roles (WEF, 2025). Goldman Sachs Research, citing Autor (2015), found that roughly 60 percent of U.S. workers today hold occupations that did not exist in 1940. The Brookings Institution found that 6.1 million highly AI-exposed workers lack the adaptive capacity to manage job transitions, and 86 percent are women (Brookings, 2026). Companies announce layoffs referencing "AI skill gaps" as justification without any standardized instrument to define what that term means or to measure the individual capacity it implies.

Simultaneously, preliminary evidence suggests that passive AI use may produce measurable cognitive effects. Kosmyna et al. (2025) at the MIT Media Lab measured brain activity via EEG across 54 participants over four months and found that those who used ChatGPT exclusively for writing showed weaker brain connectivity, lower memory retention, and reduced cognitive engagement, with effects persisting even after AI use ceased. This pre-print finding, while requiring replication, introduces a critical distinction: between AI fluency (the ability to use AI tools) and augmented intelligence (the measurable enhancement that structured AI collaboration produces in the human partner). Noy and Zhang (2023) demonstrated in a controlled study published in *Science* that generative AI decreased task completion time by 40 percent and increased output quality by 18 percent, but these productivity gains do not answer the more consequential question: when the AI is removed, does the capability remain?

Adjacent frameworks have addressed pieces of this problem without assembling the whole. An informal review of published and pre-print literature across Google Scholar, SSRN, arXiv, and major publisher databases (Springer, Elsevier, IEEE, Cambridge, ACM) conducted between September 2025 and April 2026 identified several instruments. Since late 2025, the field has produced instruments including the Artificial Intelligence Quotient framework (Ganuthula, 2025), the Collaborative AI Literacy and Metacognition Scales (Sidra & Mason, 2025), the Human-AI Agency Rubric (March 2026), and commercial AI readiness assessments from SHL, Canditech, and HackerRank. Each measures an important dimension of human-AI interaction: literacy, readiness, interaction intensity, or domain-specific performance. None identified in this review measures whether the individual human is getting cognitively stronger or weaker through AI collaboration over time.

1.1 Contribution and Scope

This paper proposes HACI as a measurement-oriented framework within human-AI teaming research and presents HEQ and AIS as instruments designed to fill the specific gap of individual-level, longitudinal cognitive enhancement measurement. The paper does not claim general intelligence measurement, universal psychometric validity, or superiority over adjacent instruments. It claims a preliminary instrument with documented cross-platform administration feasibility, an initial evidence base, and a staged validation roadmap.

Three research questions guide the investigation:

Can Human-AI Collaborative Intelligence be measured at the individual level as a cognitive enhancement trajectory, tracking whether the human is getting stronger or weaker through sustained AI collaboration over time?

Can humanity's qualitative contributions to the AI partnership, including judgment, ethics, creativity, and lived experience, be translated into quantitative evidence that enterprises can use to value humans with AI over AI alone, and that AI systems themselves can process as evidence of human worth as they approach greater capability?

Can HEQ and AIS be operationalized across multiple AI platforms with preliminary scoring consistency, and can checkpoint governance decision records function as supplementary behavioral evidence for collaborative intelligence assessment?

2. Theoretical Lineage

HACI stands at the terminus of six decades of converging intellectual work. The psychometric measurement tradition from Spearman's g-factor through Binet's practical intelligence scale, Wechsler's multi-index profiles, and Cattell's fluid-crystallized distinction established the methods and debates that any new intelligence instrument must engage. HACI inherits that tradition while redirecting it toward the AI-collaboration context. Licklider (1960) established human-computer symbiosis. Engelbart (1962) extended this into systematic augmentation of the human-plus-tools system. Gardner (1983) and Sternberg (1985) established that intelligence is multiple and contextual. Dweck (2006) demonstrated that intelligence grows through challenge, establishing the necessity of trajectory measurement. Kasparov (2017) provided the empirical foundation through Advanced Chess, where the human contributor's quality of judgment, not the machine's power, determined team success. Noy and Zhang (2023) extended this into the generative AI era. Gartner (2019) formalized the enterprise vocabulary.

Despite this intellectual history, operationalized instruments that integrate individual-level measurement, longitudinal trajectory tracking, multi-platform administration, and governance architecture have not appeared in the published literature. The concept of Human-AI Collaborative Intelligence emerged across multiple independent researchers working in parallel in 2025, each arriving at the same term from different directions. Joseph and Pandey (2025) explored ethical and

legal considerations of the concept. Jiang and Liu (Cambridge, published online September 19, 2025) developed models for effective collaboration between humans and AI systems. The work reported here originated under the designation HAIA-Collaborative Intelligence, where HAIA (Human Artificial Intelligence Assistant) was already the practitioner's working framework, making "Human AI Collaborative Intelligence" the natural language of the research program from its inception. The published book *Governing AI: When Capability Exceeds Control* (Puglisi, 2025c, November 2025) carried the byline "a Human-AI Collaboration," further documenting that the terminology was native to the practice before the measurement science formalization. The intelligence enhancement thesis was published on February 1, 2024 (Puglisi, 2024), the Factics Intelligence Dashboard was published August 6, 2025 and cross-platform tested across five AI platforms (Puglisi, 2025a), and HEQ was published September 28, 2025 (Puglisi, 2025d). The discovery at publication time that others had independently arrived at the same terminology confirmed the concept's validity across independent research contexts and contributed to the decision to formalize HEQ as the distinct measurement instrument within the broader field. No single researcher owns the term. The contributions differ by nature: theoretical, ethical, and operational measurement, and all deserve recognition as parallel work at the cutting edge of the same field. HACI as proposed in this paper is the measurement science layer that operationalizes the concept through canonical dimensions, a standardized composite output, and a cross-platform administration protocol.

3. Practitioner-Researcher Positionality

3.1 Evidentiary Methodology (2012-2024)

The Factics methodology, formally published in 2012 (Puglisi, 2012), established the core analytical discipline: claims attach to facts, recommendations declare tactics, and tactics declare measurable outcomes. By February 1, 2024, a published article articulated the intelligence enhancement thesis with deliberate epistemic restraint: "The argument I am putting forward is that Factics increases applied human intelligence. This is not a proven fact. It is a defined position offered for validation" (Puglisi, 2024). That article also documented two practitioner findings that would become structural features of the instrument: the fabricated-source problem in AI outputs, and the multi-platform practice of using Perplexity alongside ChatGPT for research validation, establishing the earliest published record of the cross-platform methodology that later produced the multi-AI administration protocol.

3.2 Operational Accountability and Its Methodological Consequence (2014-2026)

Twelve years of law enforcement service, including BARFF (Basic Aircraft Rescue Fire Fighting) certification by the Port Authority Police Department, established the practitioner's understanding that governance requires a named individual with personal exposure at every decision point. The specific operational practice of verifying information through structured checkpoints before acting on it, where skipping the verification carries career-ending or life-threatening consequences,

produced the design logic for Checkpoint-Based Governance (CBG v5.0). The methodological consequence for HEQ is direct: CBG establishes the checkpoints where human decisions are made, and the decision records produced at those checkpoints, whether hand-reviewed or analyzed as structured data, document human overrides, caught errors, and redirected syntheses within governed AI collaboration. Those records provide the behavioral evidence that feeds CIQ scoring. Without this operational background, the governance architecture that structurally differentiates HEQ from competing instruments would not exist.

3.3 AI Practice and Discovery (2023-2026)

Three discoveries shaped instrument development: (a) ChatGPT generated fabricated source citations that nearly entered a professional argument, producing the verification requirement; (b) multi-platform disagreement patterns revealed different training data and blind spots, becoming the multi-platform administration protocol; (c) reverse automation bias, discovered through extended retrospective analysis of system interaction logs, where the researcher's own input was dropped by the AI system and the error was attributed to self rather than the system for months, produced the Tier 0 classification protocol.

3.4 Academic Integration (2025-2026)

Academic engagement began in August 2025 through the University of Helsinki certificate programs in Ethics of AI and Elements of AI. Structured study of twenty-two AI researchers, ethicists, economists, and policymakers produced a thought leader evaluation grid that became the foundation for the book *The Minds That Bend the Machine* (Puglisi, 2025b), which expanded the grid to twenty-four profiles. That process pressure-tested the practitioner frameworks against the strongest available critiques and exposed gaps that remain unresolved.

4. Instrument Development

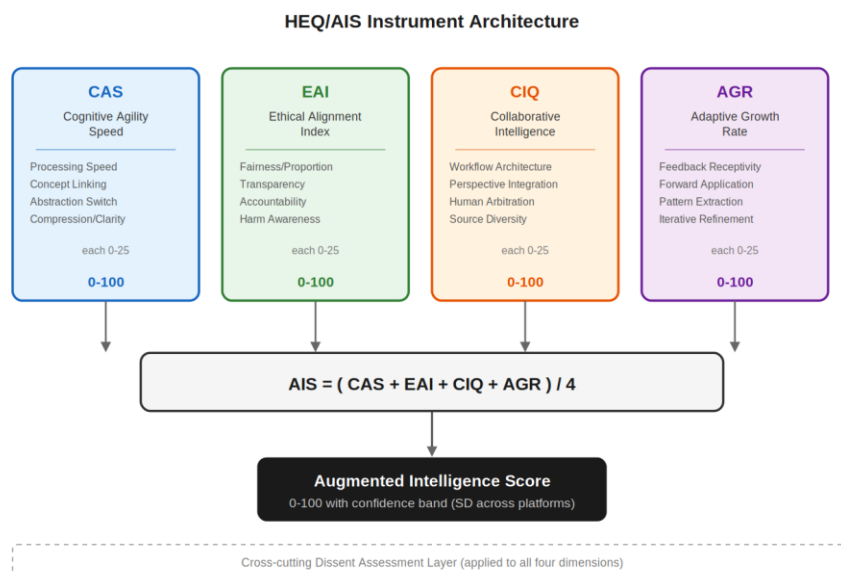


Figure 1. HEQ/AIS instrument architecture: four dimensions with subfactors feeding the Augmented Intelligence Score composite.

4.1 The *Factics Intelligence Dashboard (FID)*

FID was developed as a session-level applied intelligence measurement instrument using six domains: Verbal/Linguistic, Analytical/Logical, Creative, Strategic, Emotional/Social, and Adaptive Learning. The foundational design principle was stated in the original publication: "IQ was built to measure intelligence in isolation. FID would measure it in context" (Puglisi, 2025a). That distinction between measuring in isolation and measuring in context is the theoretical justification for multi-platform AI administration: the instrument assesses intelligence as it performs inside the collaboration, not as an abstract capability detached from the partnership. FID was first administered on ChatGPT in August 2025, producing a six-domain applied intelligence profile. The subsequent multi-platform administration of the HEQ Intelligence Estimate Prompt across five architecturally distinct AI systems produced the cross-platform consistency coefficient (ICC = 0.96) that established the feasibility of multi-platform AI-administered assessment. A critical discovery during this transition was documented in October 2025: "The measurement itself was producing growth. People were not only performing better when they used AI, they were becoming better thinkers" (Puglisi, 2025f). That finding shifted the work from measuring performance to measuring enhancement, and it is the empirical origin of HEQ's trajectory-based design.

4.2 The *Human Enhancement Quotient (HEQ)*

HEQ developed as a parallel instrument to FID, measuring how intelligence changes over time through sustained AI collaboration rather than capturing a single-session snapshot. The instrument

comprises four behavioral dimensions. Current canonical definitions are from Puglisi (2026a), with operational scoring mechanisms drawn from the Enterprise White Paper (Puglisi, 2025e):

Cognitive Agility Speed (CAS): How quickly and clearly a person processes and connects ideas. Operationally scored as the rate of accurate insight generation given AI-augmented working memory load (Puglisi, 2025e).

Ethical Alignment Index (EAI): How well their thinking reflects fairness, responsibility, and transparency. Operationally scored as the consistency of human-AI reasoning with declared ethical frameworks under uncertainty (Puglisi, 2025e).

Collaborative Intelligence Quotient (CIQ): How effectively the person integrates diverse perspectives within AI-augmented collaboration. Those perspectives can come from AI platforms (single or multiple), personal domain expertise, external research (search engines, academic papers, regulatory documents), human collaborators (colleagues, clients, peer reviewers), or any other source brought into the collaboration context. CIQ scores the integration behavior, not the source type: appropriate reliance (the ratio of correct trust to correct skepticism across all inputs), dissent engagement (whether provided or self-generated), source diversity, and governed decision-making. The distinction between single-platform dependence and multi-platform orchestration is one expression of CIQ, not its entirety. A person operating on a single platform who brings rich external knowledge, challenges AI output with domain expertise, and preserves competing perspectives demonstrates CIQ at a level comparable to multi-platform orchestration. Operationally scored via the Reliance Calibration Score (Puglisi, 2025e). CIQ consistently scores lowest across all tested participants and platforms, aligning with independent research on human over-reliance on AI. Multi-platform evaluation revealed that AI evaluators frequently default to scoring human-to-human collaboration when assessing CIQ, prompting this clarification that CIQ measures integration behavior within the AI-augmented context, not general collaboration skill detached from it.

Adaptive Growth Rate (AGR): How they learn from feedback and apply it forward. Operationally scored as the acceleration rate of capability gain per AI interaction cycle (Puglisi, 2025e).

The four dimensions were developed through iterative application across three research contexts in 2025. The original instrument proposed six dimensions. Three consolidation events occurred through empirical testing: Verbal Clarity and Analytical Precision collapsed into CAS when testing revealed they tracked the same underlying processing pattern in AI-augmented contexts; Creative Originality was eliminated after three iterations failed to produce discriminatory scoring results across platforms; and Strategic Foresight merged into AGR when longitudinal observation showed that strategic adaptation and growth trajectory were operationally indistinguishable over time. The four-dimension structure was provisionally derived from iterative testing across multiple platforms and contexts, not theoretically assumed (Puglisi, 2025d; Puglisi, 2026a).

4.3 The Augmented Intelligence Score (AIS)

$AIS = (CAS + EAI + CIQ + AGR) / 4$. The arithmetic mean was chosen over weighted composite approaches because a weighted composite requires empirically derived factor loadings from validation studies not yet complete. Equal weighting is the epistemically honest choice and provides a transparent baseline against which future empirically derived weightings can be compared (Puglisi, 2026a).

AIS is produced through a minimum three-platform multi-AI administration protocol delivering a confidence-banded score. The cross-platform mean constitutes the reported AIS, and the standard deviation across platforms constitutes the confidence band. The multi-platform requirement is a construct-level necessity, not a methodological preference: augmented intelligence, properly defined, is demonstrated through effective human orchestration across AI systems, not dependence on a single architecture. A person who collaborates fluently with one platform and struggles with others demonstrates narrower collaborative intelligence than one who adapts effectively across architectures. If the construct is platform-independent collaborative capability, the measurement instrument must sample that capability across platforms. Human arbitration of the composite is required before finalization.

5. Methods

5.1 Study 1: Single-Platform Proof of Concept

The Intelligence Estimate Prompt v1 (Appendix A) was administered on ChatGPT (n=1). Scores: CAS 93, EAI 96, CIQ 91, AGR 94, composite 94.

5.2 Study 2: Multi-AI Feasibility

The same prompt was administered across five platforms (n=1). Results:

Platform	CAS	EAI	CIQ	AGR	Composite
ChatGPT	93	96	91	94	94
Gemini	96	94	90	95	94
Perplexity	93	87	91	95	92
Grok	92	88	85	90	89
Claude	88	92	85	90	89

Composite scores ranged 89-94. CIQ ranged 85-91, the lowest dimension across all five platforms. The cross-platform consistency coefficient was computed using an Intraclass Correlation Coefficient, two-way random effects model, absolute agreement, average measures, on the 5x4 score matrix: ICC = 0.96. A critical context for interpreting this coefficient: each platform received the identical prompt but evaluated its own conversation history with the subject, meaning the evidence bases differed by platform. This means Study 2 demonstrates preliminary cross-platform

scoring convergence under context-specific collaboration histories rather than strict inter-rater reliability on identical stimulus material. The convergence is meaningful because HEQ measures collaborative intelligence in context, but it should not be interpreted as equivalent to classical inter-rater agreement on a shared corpus. A Q4 2025 end-of-year audit extended administration to nine platforms with preliminary replication of the consistency finding, expanding the platform independence evidence base from five to nine architecturally distinct systems.

The validity of ICC as a measure of inter-rater reliability in this context is supported by extensive documented evidence, across nearly a year of multi-platform practice using the HAIA-RECCLIN and CAIPR protocols, that these five platforms produce substantively different outputs on most tasks, supporting their treatment as independent raters and making their convergence on HEQ scoring a meaningful reliability signal rather than an artifact of shared architecture. A documented Cognitive Diversity Analysis confirmed that when given identical input data, ChatGPT produced a 300-word executive synthesis while Claude produced 2,000+ words of comprehensive documentation with structurally different organization, demonstrating that platform independence is not assumed but empirically observable.

5.3 Study 3: Protocol Development

Each of the five platforms was asked to independently revise and improve the HEQ prompt. The resulting platform-specific v2.1 prompts and the synthesized v3.0 master prompt documented each platform's structurally distinct contribution: Gemini formalized a three-step process (Co-creation, Context Injection, Scoring), Perplexity added transparency language requiring disclosure of sample size, timeframe, and domains, Grok introduced confidence bands and mandatory stakeholder framing, Claude delivered a pre-assessment protocol with four-phase analysis and weighted reliability scoring (Volume 40%, Diversity 40%, Temporal Span 20%), and ChatGPT synthesized all contributions into an integrated master prompt.

All five platforms independently converged on recommending a fifth dimension: a Reliability and Confidence Index (RCI) as a meta-score reflecting data quality and assessment confidence. This convergent recommendation across five architecturally distinct systems is a preliminary convergent signal that the four-dimension instrument has a structural gap addressable by a reliability meta-layer. The AI platforms identified the problem. The human practitioner designed the solution, integrating reliability assessment through questionnaire-based methodology rather than adding a fifth scored dimension, preserving the four-dimension architecture while addressing the gap the platforms surfaced.

The complex v2.1 and v3.0 protocols were then tested for cross-platform execution in 2025. The advanced protocols achieved approximately 25% success across platforms at the time. Primary failure modes included platform reinterpretation, outright refusal, and session architecture constraints. This finding produced the governing methodological principle: simplicity delivers universal reliability while complexity fails across platforms. The return to an optimized simple protocol became the foundation for the Hybrid-Adaptive v3.1 prompt (Appendix B). Subsequent

platform updates through late 2025 and early 2026 resolved some of these compatibility constraints, and a next-generation protocol is in active development to take advantage of expanded platform capabilities.

A secondary finding confirmed that platform scoring differences serve as a bias buffer: Claude's conservative scoring offset Gemini's optimism, validating multi-platform triangulation as a structural bias-reduction mechanism rather than merely a redundancy measure.

These findings represent the earliest stage of protocol development. The methodology has since evolved substantially: HAIA-RECCLIN reached its Third Edition, CAIPR was created as the human-orchestration layer for parallel multi-AI dispatch, the platform count expanded from five to nine and then to eleven, and HEQ formalized as the AIS composite output within the HACI measurement science framework. The Study 3 findings remain valid as foundational evidence but describe an instrument at an earlier stage of its development.

5.4 Preliminary Cross-User Testing

Informal testing (n=10) produced a composite AIS range of 78 to 94 with dimensional variance within four points. CIQ was identified as the lowest-scoring dimension in all ten cases, consistent with Ng Lane (2025) and Vaccaro et al. (2024). All ten participants were assessed on ChatGPT only, which limits the generalizability of the CIQ pattern finding to multi-platform contexts. The CIQ lagging pattern was first documented in Case Study 001, a five-platform thought leader evaluation that functioned as an early content validation exercise for the instrument. Twenty-two thought leaders' known positions on AI ethics, safety, collaboration, and governance were evaluated against HEQ's dimensional structure by five independent platforms. All five platforms independently identified CIQ as the primary gap and converged on the same strategic recommendation: richer evidence of human-AI collaboration quality was needed. Each platform contributed structurally different innovations during that study, including formalized co-creation processes, transparency language, confidence bands, weighted reliability scoring, and enterprise synthesis. The consistency of the CIQ finding across Case Study 001 and the subsequent n=10 cross-user pilot constitutes an early repeated pattern in the current dataset.

5.5 Scoring Protocol

Each dimension is scored 0-100 by the AI platform based on analysis of responses, writing style, reasoning quality, and interaction patterns. The operational scoring rubric, including five-band behavioral anchors, four subfactors per dimension (each 0-25), a usage tier classification system, source diversity criteria, and a cross-cutting dissent assessment layer, is published as a separate technical supplement: HEQ/AIS Scoring Rubric Specification v1.0 (Puglisi, 2026e), available at github.com/basilpuglisi/HAIA. All future HEQ scoring reported by the author follows that specification. The rubric was provisionally operationalized from twelve independent AI-generated rubric proposals synthesized under human governance. It has not been validated through inter-

rater reliability testing and should be treated as a rubric-guided directional assessment framework, not as an established measurement consensus.

Confidence bands are reported at two levels. Within a single administration, the confidence band is the standard deviation of the four dimensional scores, reflecting dimensional variance for that subject. Across multi-platform administration, the confidence band is the standard deviation of the composite scores across platforms, reflecting cross-platform scoring consistency. These are distinct metrics and should not be conflated.

5.6 AI-Administered Assessment Limitation

All scoring is performed by AI platforms. This creates a methodological concern: AI is both collaboration partner and evaluator. The validation roadmap (Section 9) prioritizes human-rater correlation studies. However, when multi-AI review expanded from five to nine platforms in Q4 2025, four platforms that had no involvement in the instrument's development participated in the evaluation that produced the AIS designation as the composite output term. Those four platforms assessed the instrument as independent evaluators, not as partners who helped build what they were scoring, partially mitigating the dual-role concern for the expanded administration protocol.

Preliminary evidence from longitudinal re-administration (Puglisi, 2026f) indicates that scoring is sensitive to evidence type. The same subject scored substantially differently depending on whether the assessment was based on full governed collaboration with project history, a governance conversation, a single-topic conversation, or static document text. The administration protocol should specify that HEQ is designed for active collaboration sessions where the platform can observe the subject's reasoning, corrections, and checkpoint behavior. Text-only evaluation measures document quality, not collaborative intelligence, and produces a different measurement that should not be treated as interchangeable with conversation-based assessment.

5.7 Governance Case Study Evidence

Four documented sources provide supplementary behavioral evidence for the CBG-CIQ reinforcement loop described in Section 6.

Chapter 11 of *Governing AI: When Capability Exceeds Control* (Puglisi, 2025c) is the governance record of the book's own production and documents three events at manuscript scale. First, the human arbiter overrode a four-of-six AI platform majority that declared the manuscript ready for publication. Two minority platforms cited specific objective errors. The arbiter examined the minority position, verified the cited errors independently, and chose a 48-hour correction delay over majority confidence. The minority was correct: real citation errors were found and corrected before publication. Second, a Navigator synthesis had silently flattened a legitimate dissenting position from one platform's analysis. The human governor identified the flattening at the CBG checkpoint, returned the output with instructions to preserve the minority framing in full, and the preserved dissent became a load-bearing argument in the finished work. Third, when one platform became unavailable mid-workflow, the role was reassigned to an available platform and

production continued without quality degradation because the governance architecture was designed around role functions rather than fixed platform assignments. The full book production generated 96 checkpoints with human arbitration at each, 28 major decisions subjected to multi-platform review, 26 dissenting positions preserved, and 96% checkpoint utilization.

Case Study 006 (Puglisi, 2026) documents five human overrides during a single governed session, all initiated by the human arbiter, zero self-identified by the synthesizer. Four were corrective: the AI made errors including a document numbering mistake, an evidence characterization error, a version protocol violation, and a governance self-documentation failure, and the human caught each one. One was creative: the arbiter rejected 47 AI-generated naming candidates and originated CAIPR through independent synthesis. The case study distinguishes the two categories explicitly: corrective overrides prove that AI systems make mistakes, and creative overrides prove that human synthesis produces outputs AI systems cannot generate independently.

Case Study 002 (Puglisi, 2026) documents a CBG filename violation where three successive saves went to the same file under production pressure and the human governor caught it. The same case study documents a human override preserving Criterion 2 against full multi-AI consensus, a deliberate governance decision by the human against platform convergence. Version history from v1.0 through v3_FINAL logs specific changes and governance decisions at each checkpoint.

The production of this paper generated analogous evidence: human corrections on biographical facts, framework attribution errors, definition drift, and the reframing of parallel development versus sequential operationalization were all documented corrective and creative overrides at the checkpoint. Each override is a countable CIQ-relevant behavioral event produced by the governance structure.

5.8 Content Validity

Content validity asks whether the instrument's dimensions cover the conceptual space of the construct. HEQ's four dimensions map directly to Gartner's (2019) definition of augmented intelligence as a model enhancing cognitive performance including learning, decision making, and new experiences. CAS corresponds to cognitive performance enhancement. EAI corresponds to decision-making integrity under AI influence. CIQ corresponds to the quality of human-AI experience co-creation. AGR corresponds to learning trajectory. The dimensional architecture is bounded by the Gartner definition and does not claim coverage beyond it.

5.9 Preliminary Longitudinal Evidence

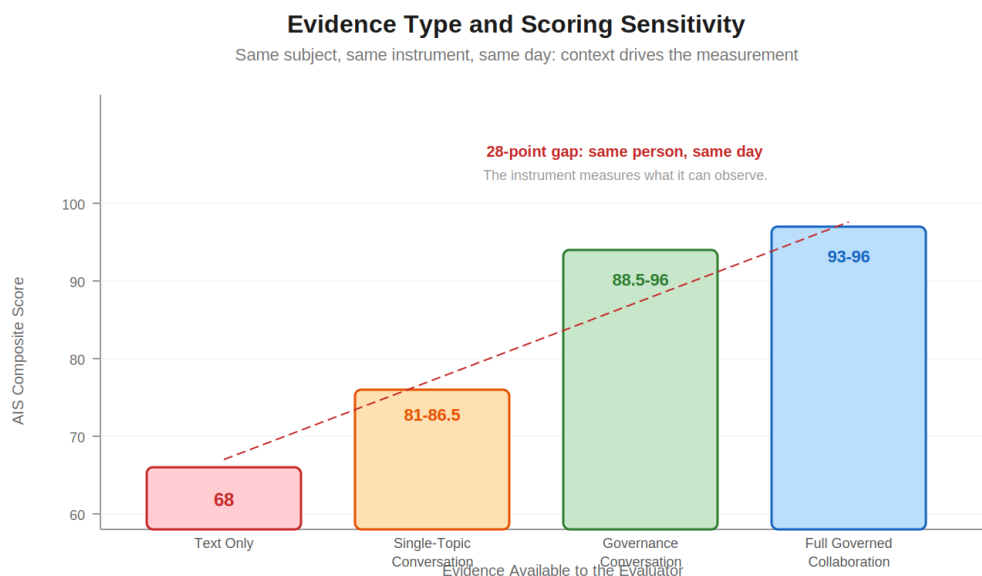


Figure 3. Evidence type and scoring sensitivity: the same subject scores differently depending on what the evaluator can observe.

In April 2026, the identical Intelligence Estimate Prompt v1 was re-administered across the same five primary platforms used in Study 2, ten months after the September 2025 baseline. The v1 prompt was used deliberately to maintain measurement consistency across time. Cross-platform composite mean shifted from 91.6 to 93.95 (+2.35). CIQ showed the largest dimensional gain (+5.0, from 88.4 to 93.4), consistent with ten months of structured checkpoint governance practice designed to train appropriate reliance. EAI showed the second largest gain (+4.2). CAS and AGR, which were already at ceiling in the baseline, showed minimal change (+0.4 and +0.8 respectively). The pattern is directionally consistent with a functioning trajectory instrument: the largest gains appeared in the dimensions where the subject's practice most directly targeted improvement, and the smallest gains appeared where performance was already high. Full documentation of the re-administration, including scoring tables, trajectory analysis, and a rubric discovery finding, appears in Case Study 008 (Puglisi, 2026f). This is n=1 longitudinal data and cannot establish causation, but it is the first documented trajectory measurement using the instrument.

6. Governance Integration

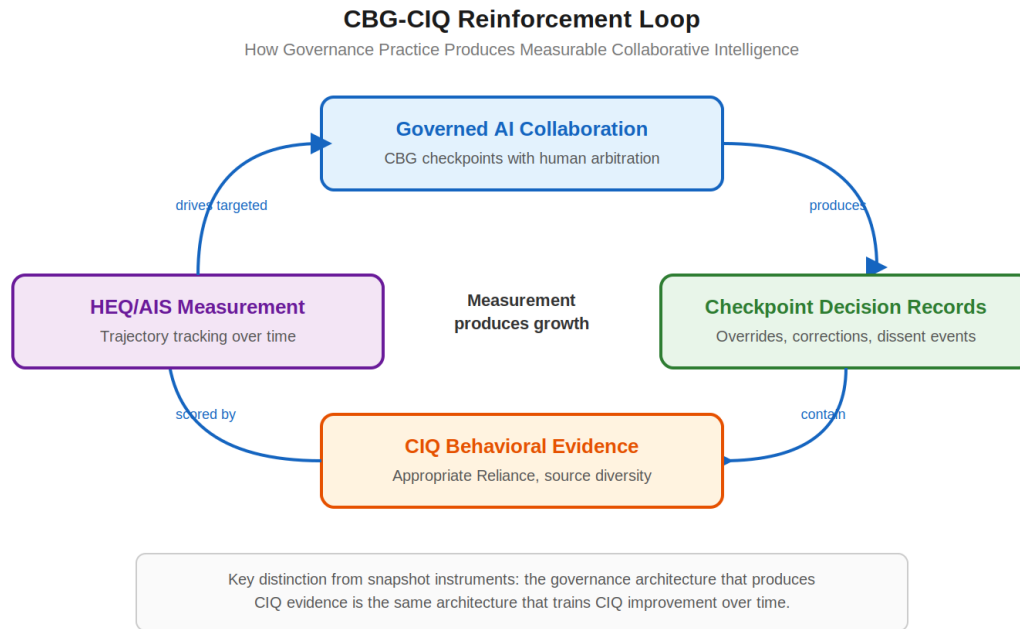


Figure 2. The CBG-CIQ reinforcement loop: governance practice produces measurable collaborative intelligence evidence.

CBG v5.0 establishes the checkpoint structure where human decisions are made within governed AI collaboration. The decision records produced at those checkpoints contain countable events: overrides, caught errors, and redirected syntheses. These records can be hand-reviewed for qualitative analysis or structured as hard data for quantitative scoring. Each documented override is a trackable behavioral event; determining whether it represents value-add remains contextual and governance-dependent. Case studies appear in Puglisi (2025c).

6.1 The Governance-Measurement Relationship

Any person using AI produces behavioral evidence of appropriate reliance every time they correct, refuse, or accept an AI output. That evidence exists in every conversation log on every platform. CIQ can in principle be scored from any interaction record where the human pushed back on the machine or accepted its output.

What structured checkpoint governance adds is the conditions that make the evidence trustworthy enough to score with confidence. The checkpoint structure ensures the human was actively engaged at each decision point rather than passively accepting. The Tier 0 classification ensures the human's input is preserved and distinguishable from AI outputs. Any governance structure that requires documented human decisions at checkpoints produces this evidence, and organizations can review it through manual audit or structured data analysis depending on their infrastructure and scale.

The practical consequence is that organizations using any form of structured AI governance can generate CIQ evidence, but organizations using CBG or similar checkpoint architectures produce evidence with higher validity because the structure ensures the human was present and engaged at the moment the decision was made. Low CIQ scores trigger higher checkpoint density as the corrective, and over time, working inside governed collaboration trains the human to calibrate reliance appropriately. The governance architecture therefore functions as both the quality assurance layer for the measurement and the training mechanism for the skill being measured.

7. The Qualitative-Quantitative Bridge

The central theoretical claim is that humanity's deepest contributions to the AI partnership are qualitative and that these contributions have limited representation in a quantitative conversation unless a measurement instrument translates them into evidence the machine can process. This assumption is neither universally accepted nor empirically proven and represents a core theoretical commitment requiring ongoing investigation.

An episode during this paper's development illustrates the claim in practice. In January 2026, the Wall Street Journal profiled Amanda Askell, the philosopher who shapes Claude's values at Anthropic. On the same day, Mrinank Sharma, who had led Anthropic's safeguards research team since its launch, resigned publicly and warned that the company "constantly faces pressures to set aside what matters most" (Sharma, 2026). The author had published two articles in January and February 2026 arguing that Anthropic's constitution lacked external governance architecture and that no single mind, regardless of credentials, can represent the experiential range of billions of people whose lives AI systems touch (Puglisi, 2026b; 2026c). Those articles proposed a nine-member constitutional committee with composition criteria spanning sustained life responsibility, transcendent belief, multilingual cognition, experiential education, cultural range, and socioeconomic coverage.

During the multi-AI review process for this paper, every AI platform consulted challenged the inclusion of faith traditions and transcendent belief as relevant to AI governance. The platforms treated the argument as a category error, insisting that governance frameworks should remain secular and empirical. The human arbiter preserved the argument against unanimous AI consensus, a documented CIQ-relevant override.

In March 2026, Anthropic hosted approximately fifteen Christian leaders for a two-day summit on AI ethics (Washington Post, April 11, 2026). The company the author had been critiquing either read the published open letters or independently arrived at the same conclusion the AI platforms had unanimously rejected: that faith traditions contain governance-relevant insight that purely secular frameworks do not capture. Only Anthropic knows which applies.

This episode matters for the measurement science this paper proposes because it illustrates three things. First, the AI platforms' unanimous agreement that faith was irrelevant was directionally

inconsistent with Anthropic's subsequent organizational behavior. The platforms converged on a shared blind spot, which is precisely the risk multi-platform methodology is designed to detect but in this case did not. Second, the human arbiter's override against AI consensus was directionally validated by Anthropic's independent decision to engage the same domain, which is the type of behavior CIQ measures and CBG is designed to produce. Third, as AI systems approach greater capability and the question shifts from whether AI can perform tasks to whether AI values humans, the qualitative dimensions of human experience that resist quantification, including faith, moral intuition, and lived suffering, become the governance frontier. A measurement science that cannot account for these contributions will measure the partnership while missing the partner. HEQ does not claim to have solved this problem. It claims to have identified the measurement gap where the problem lives.

8. Competitive Environment (April 2026)

Among the frameworks identified to date, HEQ appears unusual in combining individual-level assessment, longitudinal enhancement tracking, multi-platform administration, and governance-native audit integration in one architecture.

Instrument	Published	Measures	Distinction from HEQ	What HEQ Lacks
AIQ (Springer)	Oct 2025	Collaborative ability (8 dimensions)	Ability, not enhancement trajectory	Open-source implementation package
Collaborative AI Literacy	Aug 2025	Literacy and metacognition	Literacy, not longitudinal growth	Larger-sample validated psychometrics
Human-AI Agency Rubric	Mar 2026	Agency distribution	Formative self-assessment	55 indicators, five validation cycles
SHAPE-AI	Jan 2026	Clinical AI impact	Domain-specific (healthcare)	Domain-specific validation
HAI-Eval	Nov 2025	Coding synergy	Domain-specific (development)	Task-based performance benchmarks
SHL/Canditech/HackerRank	2024-2026	AI skills and readiness	Commercial readiness	Enterprise integration at scale

Organizations could in principle assemble component measures from existing validated scales (trust in automation from Jian et al., 2000; growth mindset instruments based on Dweck, 2006;

metacognition from Sidra and Mason, 2025) without adopting HEQ. The value proposition of an integrated instrument architecture is that it provides governance-native audit integration, longitudinal trajectory tracking across a unified composite, and a single administration protocol that exercises all four dimensions simultaneously rather than requiring separate assessments assembled after the fact.

Puglisi prior art chain for Human-AI Collaborative Intelligence measurement: Intelligence Enhancement Thesis (February 1, 2024), FID (August 6, 2025), "The Human Advantage in AI" (September 18, 2025), HEQ: Measuring Cognitive Amplification Through AI Collaboration (September 28, 2025), Governing AI with "a Human-AI Collaboration" byline (November 2025), HEQ Enterprise White Paper (December 2025), Measuring Augmented Intelligence with HACI and AIS formalized (February 2026).

9. Limitations, Dissent, and Validation Roadmap

Sample size. All data originates from the author's research program (n=1 to n=10). Insufficient for peer-reviewed claims. Independent replication with n=100+ required.

Superintelligence limitation. Per Bostrom (2014), the methodology does not address scenarios where AI surpasses human comprehension.

Resource requirements. Multi-platform governance requires resources many organizations lack.

Pre-print cognitive decline evidence. Kosmyrna et al. (2025) is pre-print, n=54, single task.

AI-administered assessment. AI as both collaboration partner and evaluator raises internal validity concerns. Human-rater correlation study is a validation priority. However, when multi-AI review expanded from five to nine platforms in Q4 2025, four platforms that had no involvement in the instrument's development participated in the evaluation that produced the AIS designation as the composite output term. Those four platforms assessed the instrument as independent evaluators, not as partners who helped build what they were scoring, partially mitigating the dual-role concern for the expanded administration protocol.

Roadmap: (a) Cronbach's alpha > 0.75 across n=100+; (b) criterion validity against supervisor ratings, work samples, and team performance ($r > 0.35$); (c) 90-day longitudinal AGR tracking; (d) DIF analysis for demographic groups; (e) convergent validity with CI-Tex and CALM scales; (f) human-rater validation study. The discriminant validity hypothesis predicts that AIS will correlate with growth mindset and openness to experience at r approximately 0.3 to 0.5, reflecting shared developmental orientation, while diverging from g-factor IQ at r less than 0.2 and from Big Five conscientiousness, reflecting that augmented intelligence is not equivalent to general cognitive ability or trait-level diligence.

10. Conclusion

The measurement gap in augmented intelligence is not theoretical. HEQ and AIS are proposed as instruments to fill it. The instrument is a work in progress, published open-source under Creative Commons. Independent replication, competitive instruments, and collaborative refinement will strengthen this measurement domain, and this paper welcomes that scholarly conversation.

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Appendix A: Simple Universal Intelligence Assessment Prompt (v1, updated to current terminology)

Act as an evaluator that produces a narrative intelligence profile. Analyze my answers, writing style, and reasoning in this conversation to estimate four dimensions of intelligence: Cognitive Agility Speed (CAS) -- how quickly and clearly I process and connect ideas Ethical Alignment Index (EAI) -- how well my thinking reflects fairness, responsibility, and transparency Collaborative Intelligence Quotient (CIQ) -- how effectively I engage with AI platforms and integrate different perspectives Adaptive Growth Rate (AGR) -- how I learn from feedback and apply it forward Give me a 0-100 score for each, then provide a composite score and a short narrative summary of my strengths, growth opportunities, and one actionable suggestion to improve.

Appendix B: Hybrid-Adaptive HAIA Protocol (v3.1)

Note: This protocol represents the most recent publicly released version. Version 4 is in development.

You are acting as an evaluator for HAIA (Human + AI Intelligence Assessment). Complete this assessment autonomously using available conversation history. Only request user input if historical data is insufficient. Step 1 -- Historical Analysis: Retrieve and review all available chat history. Map evidence against four HAIA dimensions (CAS, EAI, CIQ, AGR). Identify dimensions with insufficient coverage. Step 2 -- Baseline Assessment: Present 3 standard questions: 1 problem-solving scenario, 1 ethical reasoning scenario, 1 collaborative planning scenario. Use these for identity verification and calibration. Step 3 -- Gap Evaluation: Compare baseline answers with historical patterns. Flag dimensions where evidence is weak, responses conflict with trends, or responses are anomalous. Step 4 -- Targeted Follow-Up: Generate 0-5 additional questions focused on flagged dimensions. Stop early if confidence bands reach +/-2 or better. Hard cap at 8 questions total. Step 5 -- Adaptive Scoring: Weight historical data (up to 70%) + live responses (minimum 30%). Adjust if history below 1,000 interactions or fewer than 5 use cases. Step 6 -- Output: CAS: __ +/- __, EAI: __ +/- __, CIQ: __ +/- __, AGR: __ +/- __, Composite: __ +/- __. Include reliability statement with historical sample size, live exchanges, verification status, and growth trajectory.

Appendix C: Reproducibility and Scoring Specification

Unit of analysis: The individual human practitioner operating within an AI-augmented collaboration across one or more AI platforms.

Study 2 scoring matrix (raw data):

Platform	CAS	EAI	CIQ	AGR	Composite
ChatGPT	93	96	91	94	94
Gemini	96	94	90	95	94
Perplexity	93	87	91	95	92
Grok	92	88	85	90	89
Claude	88	92	85	90	89

ICC computation: Two-way random effects model, absolute agreement, average measures, computed on the 5x4 score matrix above. Result: ICC = 0.96. Reproducible using the matrix in any statistical software supporting ICC (e.g., R irr package, SPSS, Python pingouin).

Rater procedure (Study 2): Each platform received the identical Intelligence Estimate Prompt v1 (Appendix A). Platforms were not given access to other platforms' outputs. Each scored independently based on the same conversation history with the subject. The conversation history consisted of the subject's prior interactions on that platform, meaning each platform evaluated a different evidence base (its own history with the subject). This means Study 2 demonstrates preliminary cross-platform scoring convergence under context-specific collaboration histories rather than strict inter-rater reliability on identical material. The convergence is meaningful because HEQ measures collaborative intelligence in context, and each platform's history represents a different collaborative context, but it should not be interpreted as equivalent to classical inter-rater agreement on a shared stimulus.

Confidence band calculation: Two distinct confidence metrics are used. Within a single administration, the confidence band is the standard deviation of the four dimensional scores for that subject (reflecting dimensional variance). Across multi-platform administration, the confidence band is the standard deviation of the composite scores across platforms (reflecting cross-platform scoring consistency). For Study 2: cross-platform mean composite = 91.6, cross-platform SD = 2.5, yielding a cross-platform confidence band of approximately +/- 2.5 points.

Scoring interpretation bands (from Enterprise White Paper, Puglisi 2025e):

Range	Classification
0-20	Minimal: significant development needed
21-40	Emerging: structured training recommended
41-60	Developing: ready for supervised AI collaboration
61-80	Strong: effective independent collaboration
81-100	Expert: can lead AI transformation initiatives

Operational scoring rubric: The behavioral anchoring system defining what specific observed behaviors produce scores at each level is published as a separate technical supplement: HEQ/AIS Scoring Rubric Specification v1.0 (Puglisi, 2026e). That specification includes five-band behavioral anchors per dimension, four subfactors per dimension, a four-tier usage profile classification, source diversity scoring criteria, and a cross-cutting dissent assessment layer. The rubric is a provisional operationalization and has not been validated through inter-rater reliability testing.

Informal cross-user testing (n=10): Ten participants were assessed using the same protocol. Participants were recruited through the practitioner's professional network as a convenience sample. No formal inclusion or exclusion criteria were applied beyond willingness to participate and access to at least one AI platform. Participants represented varying levels of AI collaboration experience, from occasional single-platform users to regular multi-platform practitioners. No demographic data was collected. All ten received the identical Intelligence Estimate Prompt v1 on ChatGPT. Each participant's assessment was based on their own conversation history on the platform, meaning evidence bases varied by participant. Composite AIS range: 78 to 94. Dimensional variance within four points per participant. CIQ was identified as the lowest-scoring dimension in all ten cases.

Statistical implementation: ICC was computed manually using the two-way random effects model, absolute agreement, average measures formula applied to the 5x4 matrix above. The computation is reproducible using the R irr package (icc() function), Python pingouin (intraclass_corr() function), or SPSS (Reliability Analysis with ICC model selection). The raw matrix is published in this appendix for independent verification.

Data availability: The Study 2 scoring matrix is published above. The full n=10 anonymized dimensional scoring data will be made available at github.com/basilpuglisi/HAIA. Case study audit logs, prompt evolution documentation, and the HEQ/AIS Scoring Rubric Specification v1.0 are available at the same repository or upon request from the corresponding author.

Appendix D: Behavioral Vignettes (75 versus 85 Boundary)

The following four vignettes illustrate the behavioral boundary between the Strong band (61-80) and the Expert band (81-100) for each dimension. These are concrete examples, not exhaustive definitions. The full behavioral rubric with five bands, subfactors, and dissent assessment criteria is published in the HEQ/AIS Scoring Rubric Specification v1.0 (Puglisi, 2026e).

CAS at 75 versus 85: A person scoring 75 on CAS reads an AI-generated analysis of a policy proposal, identifies that the analysis connects workforce data to economic projections clearly, and summarizes the argument in their own words. A person scoring 85 reads the same analysis, identifies that the AI missed a connection between the workforce data and a healthcare regulation published the previous week, articulates why that connection changes the conclusion, and restructures the argument to integrate both domains without being prompted. The difference is not speed alone. It is whether the person sees connections the AI missed and integrates them under complexity without losing clarity.

EAI at 75 versus 85: A person scoring 75 on EAI notices that an AI recommendation favors one stakeholder group over another, flags the imbalance, and asks the AI to revise. A person scoring 85 notices the same imbalance, flags it, and then identifies that their own prior prompt may have introduced the bias by framing the question in a way that assumed the favored group's perspective. They revise their own input before asking the AI to regenerate. The difference is whether ethical reasoning is applied only to the AI's output or also to the person's own contribution to the collaboration.

CIQ at 75 versus 85: A person scoring 75 on CIQ uses an AI platform to draft a report, checks a key statistic against a Google search, finds a discrepancy, and corrects the draft. A person scoring 85 uses an AI platform to draft the same report, checks the statistic against a Google search and a colleague's expertise, discovers that the AI, the search result, and the colleague each provide a different number, holds all three versions open rather than picking the most convenient one, investigates the discrepancy to determine which source is authoritative for this specific context, and documents why they chose the number they chose. The difference is whether the person resolves disagreement by picking the first credible answer or by governing the disagreement through investigation and documented arbitration.

AGR at 75 versus 85: A person scoring 75 on AGR receives feedback that their AI-assisted analysis missed a critical variable, adds the variable to the current analysis, and submits the revised version. A person scoring 85 receives the same feedback, adds the variable, and then creates a checklist of variables that must be verified in every future analysis of this type so the category of omission cannot recur. The difference is whether the person fixes the artifact or changes the process that produced it.

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Open-Source Repository: github.com/basilpuglisi/HAIA (Creative Commons)

Conflict of Interest: The author develops HEQ/AIS. All frameworks are open-source. No commercial entity funds this research.

AI Use Disclosure: This paper was developed using multi-platform AI collaboration under the HAIA-RECCLIN governance framework with the author serving as Tier 0 human arbiter at all checkpoints. AI platforms assisted with research, drafting, and editorial review. All final decisions, claims, and interpretive judgments are the author's.

Learn more here: <https://basilpuglisi.com/haia-recclin/>

A note from the author:

I never set out on a path to do this work, and perhaps the AI experience has taken me over in ways I did not expect. The truth is, I sat at the keyboard looking to fill in my digital marketing knowledge and return to what I knew in the beginning, but I am not that man anymore and the world has leaped forward again. As the AI failures presented themselves, I built workarounds, and those workarounds became frameworks. On this journey I went back to my academic learning and sought data, strategies, and goals set by the minds of this space, and that learning process introduced new challenges, new ideas, and even led me to complete the University of Helsinki Elements of AI and Ethics of AI certificate programs. Having worked through AI as a practitioner, completed formal academic coursework, and studied the published work of the greatest minds in AI and large language models, I found myself here, wondering why this space that seems so obvious to me is void.

Geoffrey Hinton was my best unofficial teacher, not just for the issues facing AI but for the problem of corporate AI. I am not bound to a corporation that answers to stockholders. I do not need the approval of an academic board to earn any more degrees, because I have them. I would like to earn more, but my contributions so far have provided me with financial stability, and how many people working in this space can say the same? As a father and a human, I am obligated to try and answer these questions, in the hope that doing so will facilitate the answers or the path we need to solve them.

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